

Review

Applications of zwitterionic polymers

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ABSTRACT

Zwitterionic polymers are characterized with equal anion and cation groups on the molecular chains, which make them highly hydrophilic and antifouling. They can resist nonspecific protein adsorption, bacterial adhesion, and biofilm formation. Therefore, they have great potential to be applied in a wide range of biological and medical related fields, such as antifouling coatings of biomedical implants, blood contacted sensor and drug delivery in vivo, separation membrane and marine coatings. The review mainly focuses on the progress of those applications of zwitterionic polymers on the molecular level. Problems existed in these applications are also discussed and the development of in future is prospected.

1. Introduction

Biofouling, such as nonspecific protein adsorption, microorganism adhesion, and biofilm formation, is of the most important phenomena and a huge problem from biosensors, implanted medical devices, drug delivery systems, separation membrane to enormous ship hulls. It causes many problems like haemolysis, thrombosis, anticoagulation-related haemorrhage, immune responses, infection, and tissue overgrowth in implanted devices [1,2]. Biofouling can also increase the voyage resistance by 60% and decreases the voyage rate by 10%. One of the crucial factors affecting fouling is the surface properties because they determine the interaction between fouling substance and the surface.

To address the fouling problem, various hydrophilic polymers, including poly(vinyl alcohol), poly(*N*-vinylpyrrolidone), poly(2-oxazoline) and poly(ethylene glycol) (PEG), and zwitterionic polymers, have been developed to construct the hydrophilic surface, and these polymers can evidently reduce the biofouling on the surface. It should be noted that PEG is a biocompatible polymer and it becomes a gold standard for unfouling polymers [3] because of its high hydrophilicity, non-toxicity, and anti-protein-fouling [4]. However, PEG has poor stability as PEG macromolecular chains can rapidly autoxidize and degrade during storage and handling at room temperature, especially by transition metal ions, which exists in most biological related solution [5–7]. The autoxidation mechanism is a free radical mechanism initiated by minor amounts of free radicals present or catalyzed by metal salts, e.g. copper sulfate [8]. Peroxides and hydroperoxides are the

primary oxidation products followed by formation of carbonyl compounds, including ethoxylated formats, formaldehyde, ethoxylated aldehydes, and acetaldehyde, as secondary oxidation products [7,9,10]. It is shown that oligo(ethylene glycol)-terminated SAMs decompose within a week at 45 °C or after 1 month at 20 °C [11], and the decomposition is greatly accelerated by the direct contact between ethylene glycol segments and a catalyst (e.g., gold), oxygen and higher temperatures [12]. PEG brushes lost their antifouling capacity when the temperature rises to 35 °C [13]. Furthermore, it is frequently reported that PEG induced antibodies after repeated injection and PEGylation reduces the bioactivity of conjugated therapeutic proteins, antibodies, and enzymes [14,15]. In addition, PEG cannot be metabolized naturally. These drawbacks greatly limit the application of PEG. Therefore, it is urgently to seek alternative non-fouling material other than PEG.

Zwitterionic polymers are just perfect alternatives for PEG. As compared with amphiphilicity of PEG, zwitterionic polymers are super hydrophilic due to the presence of abundant ions and subsequent strong hydration layer.

Zwitterionic polymers refer to a family of materials that have the same number of cations and anions along their polymer chains. Typical cations are quaternized ammonium, and zwitterionic groups can be classified into sulfobetaine (SB), carboxybetaine (CB), phosphorylcholine (PC) according to anions. Specially, zwitterionic group is SB when anions are sulfonates, CB when anions are carboxylates, and PC when anions are phosphonates. Hydrophilicity of anionic groups decreases with increasing acidity in the order: $\text{COO}^- > (\text{RO})_2\text{P}(=\text{O})\text{O}^- > \text{SO}_3^-$ [16]. SB based polymers are most promising to be

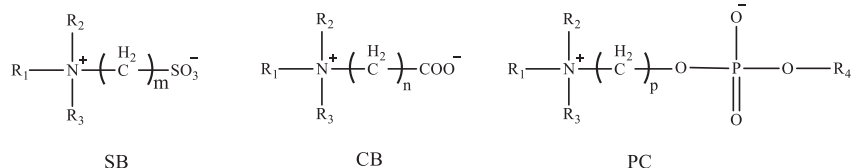
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industrialized because it is easy to prepare SB monomers, and some of them are commercially available. CB-based materials have attracted great interests for their advantages over other zwitterionic materials, including their super antifouling properties [17], good biocompatibility [18] and functionality. PC based polymers also have excellent bio and blood compatibilities, but their high production cost hinders its wide application.

As compared with other moieties with no charges, these materials are characterized with both high dipole moments and highly charged groups [19]. Due to the presence of so many cations and anions on the macromolecular chains, zwitterionic polymers are super hydrophilicity, while other polymers such as PEG are amphiphilic with both hydrophilic and hydrophobic properties [3]. It is reported that CB based zwitterions have better unfouling properties than PEG, and their self-assembled micelles have greater stability than PEG analogs during lyophilization [20,21]. The superior antifouling properties of zwitterionic materials in complicated environment were derived from their strong interaction with water via ionic solvation, while the antifouling capacity of PEG was dependent on hydrogen bond with water [22].

Besides their excellent antifouling capacity, zwitterionic materials have various other merits such as ease of functionalization and design flexibility [23]. It has many choices of zwitterions to be chosen from. Especially, as far as PCB is concerned, it can be functionalized by the –COOH groups through the 1-ethyl-3-(3-dimethylamino) propyl carbodiimide (EDC) and *N*-hydroxysuccinimide (NHS) chemistry. Moreover, PCB have many derivatives and their polymer backbones can be (meth)acrylate or (meth)acrylamide [24]. There are various choices of substitution groups of quaternary ammonium, and number (*n*) of carbon spacers between carboxylate and quaternary ammonium groups [24–26]. Thus, zwitterionic materials have a wide variety of applications in antifouling blood contacted sensors, drug delivery, etc. Besides these biomedical applications, they are also very useful in chemical separation membrane and marine coating.



Just as any other functional polymers, zwitterionic polymers can be synthesized by two different approaches: (1) direct polymerization of zwitterionic monomers or (2) post zwitterionization of reactive polymers. Evidently, polymerization from zwitterionic monomers is the most straightforward strategy to obtain zwitterionic polymers, and can achieve polymers with 100% zwitterion functionality. Most frequently, they are prepared by free radical polymerization [27]. But it's difficult to synthesize well-defined polymers or polymers with complex architectures (e.g., block copolymers and star copolymers) through free radical polymerization. Controlled radical polymerization, such as atom transfer radical polymerization and particularly the reversible addition fragmentation transfer method, is a convenient strategy to obtain polybetaines with defined end groups and molecular weight, and block and star copolymers [28–30]. As compared with direct polymerization, post zwitterionation is easy to perform and it is allowed to prepare reactive polymers with adjustable chemical structures. But the steric hindrance from neighboring group can have some adverse effects on reaction kinetics and yield can seldom be 100%.

This review will be focus on development and progress of the applications of zwitterionic polymers, such as PSB, PCB and PPC, in biomedicine, separation membrane and marine coating. Examples of the methods to construct the antifouling surfaces or nanoparticles for these

applications will be given and discussed from molecular level. Problems existed in these applications are also discussed and their future is also prospected.

2. Applications

Due to their favorable unfouling capacities, zwitterionic materials have numerous applications, including biosensors, drug delivery, cell preservation, as well as chemical separation and marine coatings.

2.1. Blood contacted sensor

2.1.1. Application of PSB in blood contacted sensor

PEG and poly(sulfobetaine methacrylate) (PSBMA) were used to modify the surface of the nanostructured portable imaging SPR chip and the unfouling properties have been compared [31]. The zwitterionic polymer surface showed the best antifouling properties. Well-controllable electrochemically mediated atom transfer radical polymerization (eATRP) was very useful to prepare ultrathin PSBMA layers on electrode surfaces and minimize the biofouling impact on the implantable neutral electrodes [32,33]. The resulting polymer brush coatings repel over 99% nonspecific protein adsorption and long-time stability, which is favorable for the application of sensors. The sensitivity of the evaluated sensor was maintained at 94% after 15 days. The overall sensitivity drift was 7% higher than that of the polyurethane-coated ones and commercially available glucose biosensors.

Although PSB brushes have been successfully employed to construct anti-fouling surfaces for biosensors as discussed above, their biosensing application has been largely restrained due to the intrinsic lack of functional groups [34]. So far, three ways are employed to introduce functional groups to PSB brushes.

The first one is copolymerization with nonzwitterionic monomers with functional groups. Wen et al. introduce the epoxide functional

group into SB copolymer by copolymerizing of sulfobetaine methacrylate (SBMA) with nonzwitterionic glycidyl methacrylate. The resulting copolymer was grafted onto surface-enhanced Raman scattering (SERS) substrate through covalent bond by the reaction between the epoxide group on the copolymer and the –SH group on SERS substrate surface [35]. Malachite green (MG) was easily trapped on copolymer-grafted Ag nanocube and enhanced the SERS signals. The drawback is that the introduction of (10 or 50%) nonzwitterionic epoxide units diminished the antifouling performance.

Zuilhof et al. develop another approach by postmodification of fully zwitterionic sulfobetaine brushes through top exchange reactions of the remaining terminal halogen substituent after ATRP [36,37]. The problem of this approach is that it is only applicable to relatively thin brushes (20–25 nm) since the end groups in thicker brushes are buried inside the brush [38], and unavailable for functionalization. Another shortcoming of this method is that number of introduced functional groups is very limited.

To overcome the limitations outlined above, Zuilhof et al. develop a new method that allows for efficient three dimensional polymer brush functionalization without compromising the zwitterionic character and antifouling properties. They synthesized a novel SB-based zwitterionic monomer equipped with a clickable azide moiety [39].

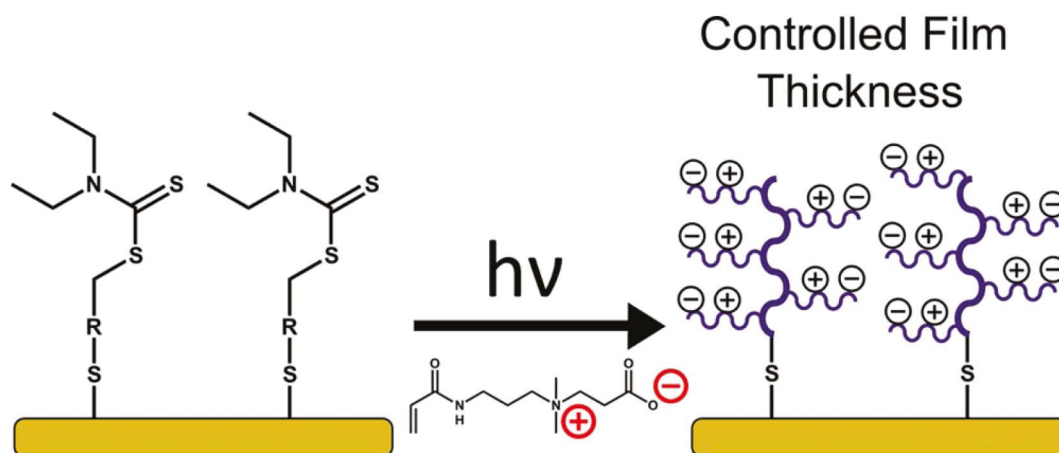


Fig. 1. Preparation of PCB film with controlled thickness [43].

Copolymerization of this monomer with SBMA results in highly anti-fouling surface coatings with a large yet tunable number of clickable groups present throughout the entire brush. Subsequent modification with a recognition unit of interest yields a highly efficient sensing platform for selective binding with retained excellent antifouling properties.

2.1.2. Application of PCB in blood contacted sensor

Among zwitterions, PCB is most commonly used to modify sensor, due to the excellent anti-fouling property and the ease of functionalization and design flexibility.

It can be applied for glucose biosensors. Poly(carboxybetaine methacrylate) (PCBMA) based hydrogels were employed to improve the biofouling capacity of implantable electrochemical glucose biosensors in complex media [40]. The resulting glucose sensor coated with PCBMA hydrogels via the chemical method has a very high sensitivity and good linearity in response to glucose in PBS, 10%, 50%, and 100% human blood serum for > 12 days.

As mentioned before, besides antifouling properties, PCB has additional advantages. It has abundant –COOH groups which can be used to decorate with functional groups or bioactive molecules. The gold surface grafted with PCBMA can resist the nonspecific adsorption of fibrinogen, lysozyme, and human chorionic gonadotropin (hCG). After binding *anti*-hCG, the surface can be used to detect hCG as it can specifically absorb hCG while retaining a high protein unfouling capacity [41]. In another case, the polymer brushes of poly(carboxybetaine acrylamide) (PCBAA) and poly[oligo(ethyleneglycol) methacrylate] (POEGMA) were grown on gold-coated glass chips for SPR spectroscopy [42]. CB in the top layer can be used for the immobilization of NeutrAvidin antibody and helps achieve a high level of attachment of targeted protein. A prototype biosensor was able to detect a targeted protein in undiluted human blood plasma.

It should be noted that surface-initiated photoiniferter-mediated polymerization is an effective method to prepare PCB thin films with even and controllable thickness (Fig. 1) [43]. Subsequent immobilization of antibody enables the film to detect an antigen (antibody to thyroid stimulating hormone) as low as 1 ng/mL from undiluted human plasma. Besides the sensitivity, it has merits of controllability and reproductively.

Especially, the PCBAA modified surface can be used for cancer detection. It is known that the level of activated leukocyte cell adhesion molecule (ALCAM) of various carcinomas is higher than the normal level of ALCAM in blood serum [44]. The mobilization of ALCAM on the surface enables materials specific detection of ALCAM. In order to achieve sensitive detection in blood plasma, the amount of mobilized ALCAM should be high enough, and its bioactivity should be retained. At the meantime, the surface should be highly antifouling. Jiang et al.

grafted PCBAA onto silica nanoparticles via atom transfer radical polymerization (ATRP) [45]. The resulting nanoparticles can be stable for over 72 h in both negative and positive protein solutions. PCBAA-modified silica nanoparticles can be easily functionalized with ALCAM for diagnostics of cancerous cell in complex media. But no data on detection and sensitivity were provided. In another study, they functionalized surfaces of label-free suspended microchannel resonators with (3,4-dihydroxy-L-phenylalanine)₂-PCBMA₂ [46]. Combined with suspended microchannel resonators, it can detect ALCAM in undiluted serum with a limit of detection of 10 ng/mL. But the applicability of this assay to clinical cancer biomarker screening requires further improvement in sensitivity and device design.

2.1.3. Application of PPC in blood contacted sensor

Poly(2-methacryloyloxyethyl phosphorylcholine) (PMPC) is another important zwitterionic polymer which can be applied for sensor modification. For example, poly-[(propargyl methacrylate)-*ran*-(2-methacryloyloxyethyl phosphorylcholine)] (PPgMAMPC) was synthesized by reversible addition fragmentation transfer polymerization [47]. As shown in Fig. 2, the alkyne moiety of propargyl methacrylate can be used to introduce azide-containing probe via a click reaction for detecting target molecules in diagnostic and related applications. Thiol-terminated PPgMAMPC-SH, obtained by aminolysis of PPgMAMPC, was immobilized on a gold-coated SPR chip. Azide-containing biotin and peptide nucleic acid were then bound on the alkyne-containing copolymer via CuAAC. Au-PPgMA65MPC35-PNA can be used as a sensor for DNA detection, and detected the highest amount of complementary DNA sequence, with 71% hybridization efficiency, and a reasonable degree of mismatch discrimination (58% MD) between the fully complementary DNA and single-mismatched DNA.

It should be noted that PMPC can be used to systematically test C-reactive protein (CRP) content using a localized SPR sensor without any further biofunctionalization as it can specifically recognize CRP [48]. Takeuchi et al. prepared poly(2-methacryloyloxyethyl phosphorylcholine)-grafted gold nanoparticles (AuNPs) by surface-initiated atom transfer radical polymerization (ATRP) of 2-methacryloyloxyethyl phosphorylcholine with AuNPs-undecyl-Br as initiator seed particles [49]. This nanosensing system was effective for detecting CRP concentrations in a human serum solution diluted to 1% (w/w) with a detection limit of ~ 50 ng/mL and negligible nonspecific adsorption of other proteins. This level of sensitivity is sufficient for application in clinical diagnostics as highly sensitive CRP assay. The nanosensing system has several merits, such as low preparation cost, avoiding complicated bioconjugation and the need for expensive instruments, no necessary complex pretreatments, and good stability. Therefore, this nanosensing system is very useful and has great potential to be applied.

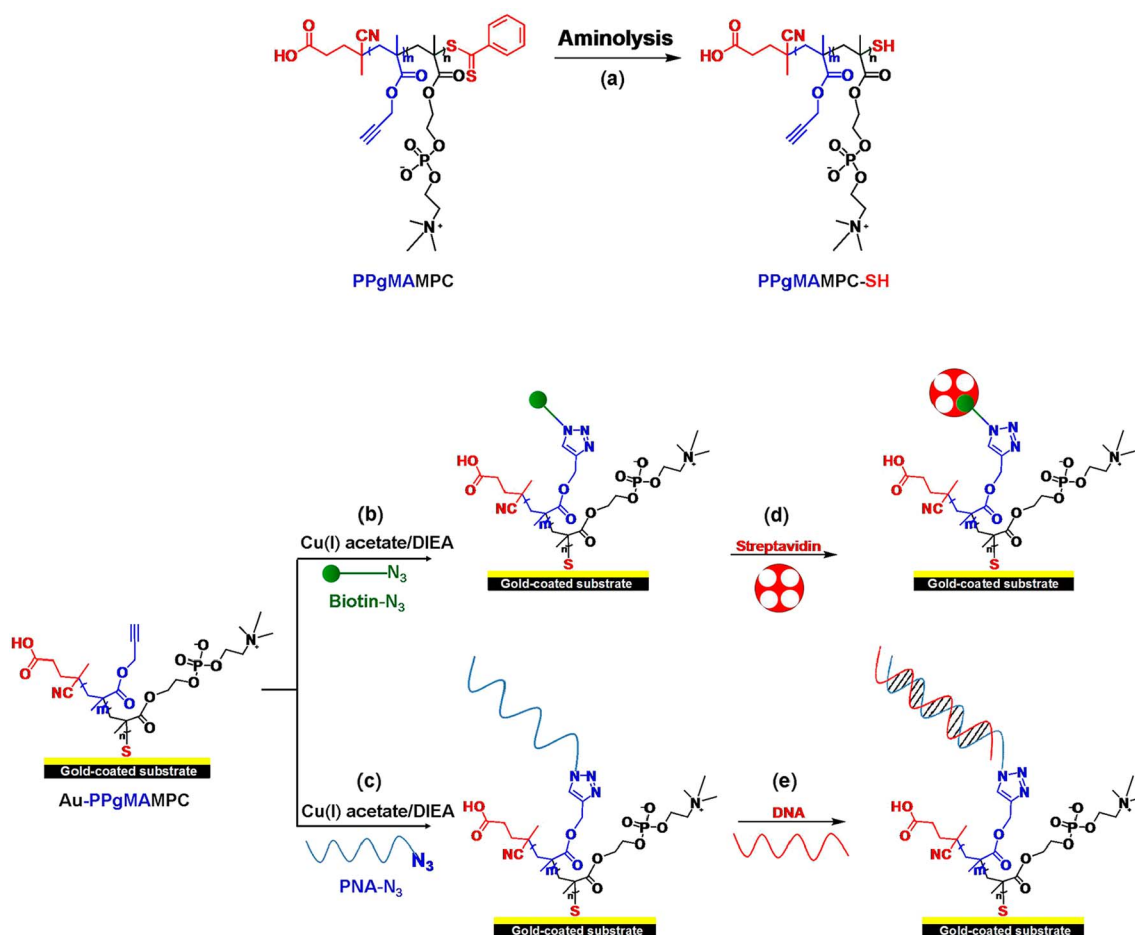


Fig. 2. Schematic diagram of (a) preparation of PPgMAMPC-SH, (b, c) immobilization of biotin-N₃ and PNA-N₃ on PPgMAMPC-modified SPR chip, and (d, e) specific binding between conjugated probe and target molecule [47].

2.2. Drug delivery

2.2.1. Increasing the circulation time and stability of drug carrier

Another significant application of zwitterionic materials is drug carrier. Because of the excellent unfouling capacity, zwitterionic polymers can avoid quick recognition by immune system and increase circulation time of nanoparticles. As far as anti-tumor drugs are concerned, the nanoparticles can passively accumulate in solid tumors due to the enhanced permeability and retention effect [50].

Generally, drug can be encapsulated into polymer micelles and nanogels or linked to polymer chains by covalent bond for drug delivery. For example, Chen et al. synthesized polymer–drug conjugates based on PMPC and camptothecin by a one-pot ATRP–“click” conjugation strategy [51]. Drug loading can be achieved to 14% and excellent solubility in pure water (> 250 mg/mL) can be retained. The linkage between the PMPC backbone and the pendant drug is highly important for controlled drug release.

Though PEG can prolong circulation time of drug-carried nanoparticles, it is very sensitive to oxidation, and easy to lose activity when the temperature rises to 37 °C. PEG also reduces the protein activity, and induces PEG-based antigen due to its amphiphilic nature. In contrast, zwitterionic polymers are more thermal and chemical stable, and they do not either induce antigen or reduce the protein activity. Gong et al. have demonstrated that micelles of random copolymer based on PC zwitterions show prolonged circulation time, whereas solely positively or negatively charged nanoparticles are easily cleared by the reticuloendothelial system [52]. The in vivo blood circulation half-life of the polymeric micelles in New Zealand Rabbits had been increased from 0.55 h to 90.5 h. Jiang and co-workers have demonstrated that the

conjugated zwitterionic polymers increase the circulation time of toxin-neutralizing butyrylcholinesterase without inducing immune antibody production as PEG [53]. This is very impressive and significant for the repeated injection of drug. Their work also shows that the biological activity of PCBMA modified enzyme α -chymotrypsin (CT), affinity for substrate and thermal stability are much better than that of PEG decorated CT [54].

Sometimes, other functional polymers have been employed to copolymerize with betaine monomers to introduce new functionality and achieve better curative effect. For example, pH-responsive poly(2-(diisopropylamino)ethyl methacrylate) (PDPA) has been copolymerized with PSBMA to incorporate new functionality [55]. The curcumin-loaded nanoparticles exhibited pH-responsive release behaviors in vivo due to the presence of PDPA segment. They were stable with slow release rate at simulating gastric fluid, whereas in simulating intestinal fluid, had increased drug release rate.

2.2.2. Enhancement of cellular internalization

The nanoparticles as drug carriers are expected to be actively internalized by tumor cells in an efficient manner in order to achieve more effective therapeutic response [56]. Unfortunately, the prolonged circulation time of PEGylated nanoparticles and PSB and PCB carried nanoparticle always results in inefficient cellular internalization [57].

It is reported that tumor targeting group, such as RGD ligand, can promote the tumoral cell internalization of zwitterionic nanoparticles by receptor-mediated endocytosis [58]. Recently, researchers are interested in charge-reversal micelles to address this problem. Specifically, the surface charge of charge-reversal micelles can switch from negative status or electric neutral status to positive status by pH-

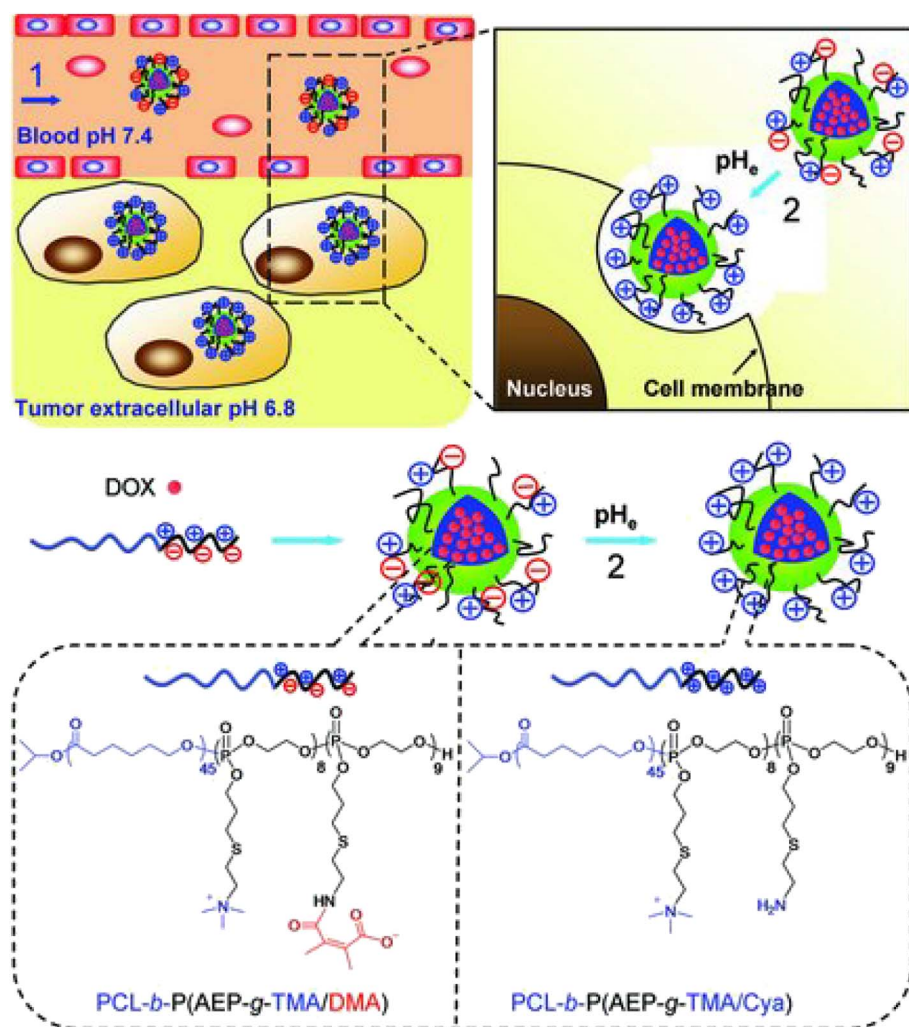


Fig. 3. Schematic illustration of doxorubicin (DOX)-loaded zwitterionic polymer-based nanoparticles and the changing of surface charge property in response to the tumor acidity (pH_e). 1) Amphiphilic zwitterionic block copolymer PCL-*b*-P(AEP-*g*-TMA/DMA) self-assembles into nanoparticles in aqueous solution with DOX encapsulation. During circulation in blood, the nanoparticles show prolonged circulation time and can leak into tumor sites through the EPR effect. 2) Responding to the pH_e, the zwitterionic polymer diminishes its anionic part, forming PCL-*b*-P(AEP-*g*-TMA/Cya), and the formed nanoparticles are activated to be positively charged and become recognizable by tumor cells [60].

sensitive bonds or protonization. The resulting positively charged nanoparticles can enter cells relatively easily and promote cellular internalization due to their greater affinity for negatively charged cell membrane [59]. Wang et al. introduced a pH (pH_e ~ 6.8) sensitive cleavable amide linkages as the anionic part of the zwitterionic polymer and developed a charge tunable nanoparticle [60]. The nanoparticles not only demonstrate prolonged circulation time, but also promote the tumoral cell internalization in vivo while they switch to being positively charged at pH_e, which in turn result in enhanced inhibition of tumor growth (Fig. 3). However, it has been reported that evident hydrolysis of these pH-sensitive bonds, e.g., β -carboxyl amide bonds, takes place at pH 7.4, resulting in premature drug release and systemic toxicity [61,62]. Moreover, pH-sensitive bonds are unstable during long-term storage under slight moist conditions. It is another problem that hinders the application of these polymeric micelles. In contrast, the protonization induced charge reversal can avoid these problems. pH-responsive alkoxyphenyl acylsulfonamide ligands has been employed to enhance the particle uptake by Rotello et al. [63]. As indicated in Fig. 4, alkoxyphenyl acylsulfonamide ligands can reversibly become cationic at tumor pH due to the protonation, and increase tumor-selective cellular uptake and cytotoxicity on HeLa cells. Our group also prepared charge-reversal hybrid micelles by simple co-assembly of cationic (carboxyl-functionalized) and anionic (amino-functionalized) copolymers based on poly(ϵ -caprolactone)-*block*-poly(butylene fumarate)-*block*-poly(ϵ -caprolactone) (PCL-*b*-PBF-*b*-PCL) [61,64]. The surface charges of hybrid micelles reversed rapidly from negative to positive at isoelectric point via protonation of surface carboxyl and amino groups.

The Nile red-loaded hybrid micelle is stable under physiological environment, but exhibits a quick cargo-release behavior under acidic tumor environment (Fig. 5). Besides charge-reversal ability, these micelles have significant advantage of biodegradability as their backbones are biodegradable aliphatic polyesters.

2.2.3. Biodegradable drug delivery systems

As compared with zwitterions based nondegradable drug delivery systems, there are much fewer reports focusing on degradable ones. Luo et al. have performed some significant work on aliphatic polyester based zwitterions, mainly SB, with desirable biodegradability. They synthesized polycaprolactone (PCL) derived zwitterionic polymers by ring opening polymerization of ϵ -caprolactone with an SB functional initiator, or through ring opening polymerization/ATRP [65,66]. The former strategy is simple and easy to realize, and the problem lies in the fact that the number of zwitterions introduced is very limited. They further introduced the pH-sensitive tertiary amine containing segment into the backbone of the copolymers to realized pH-triggered drug release [66–68]. As compared with linear polymer, pH sensitivity of star polymers are much better, and their anti-tumor effect is comparable with that of doxorubicin HCl, but the toxicity is substantially lower [67].

Jiang's group conducts important work on CB based biodegradable drug carriers. They have synthesized poly(lactic-*co*-glycolic acid) (PLGA)-PCB block copolymers (PLGA-*b*-PCB) by conjugation of PLGA terminated by COOH groups with PCB terminated by NH₂ groups (Fig. 1) [69]. The nanoparticles (NPs) self-assembled from the block

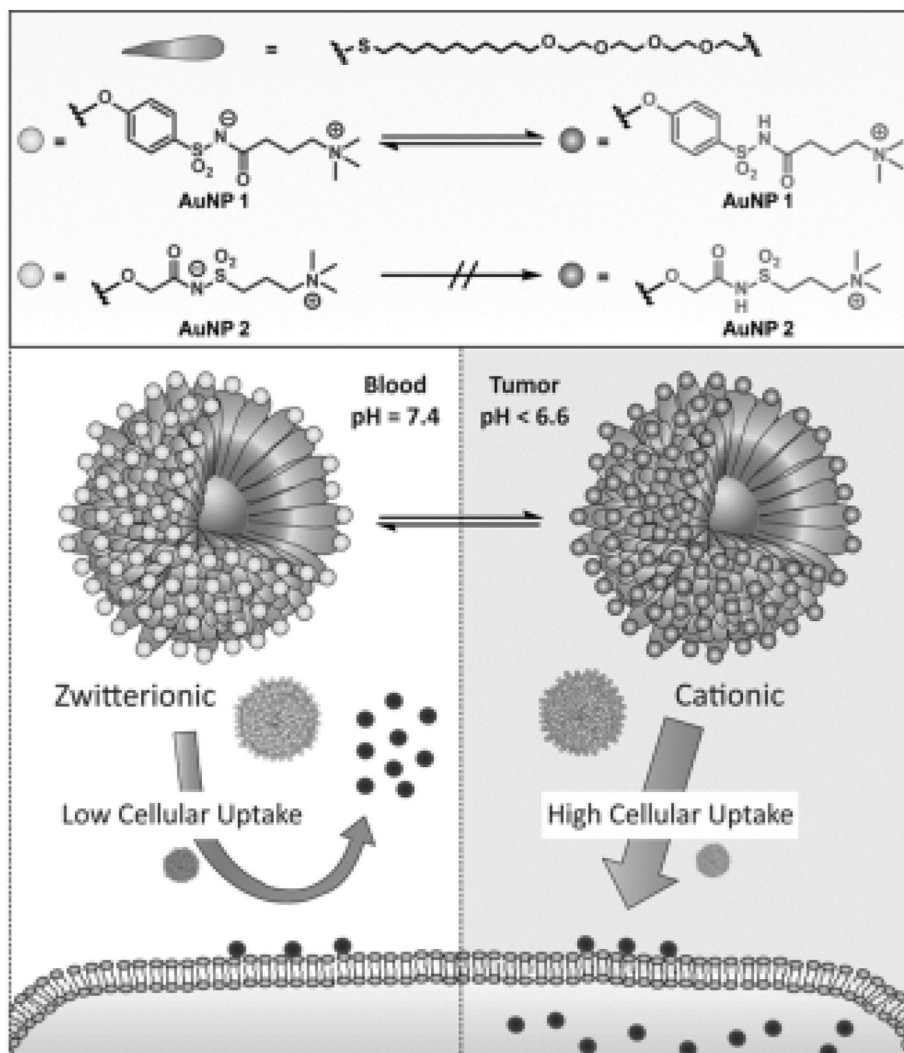


Fig. 4. Chemical structure of the monolayer-protected gold nanoparticles (AuNPs) and our strategy for a pH-responsive delivery system into tumors [63].

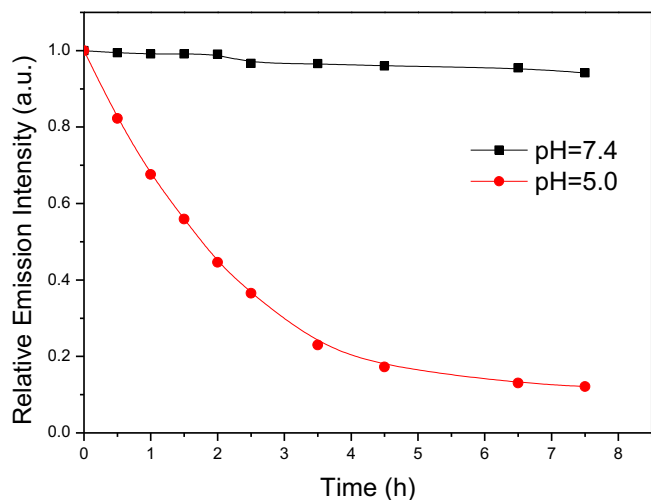


Fig. 5. Time dependence of the relative emission intensity of Nile red-loaded hybrid micelles at different pH values [64]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

copolymers possessed good stability during free-drying without addition of cryoprotectant [69], whereas 10% sucrose is needed as cryoprotectant to stabilize PEGylated NPs [70]. The obtained NPs can be easily functionalized with amine-terminated molecules which are useful

for targeting purposes. A green fluorescent dye called NBD (*N*-(7-nitrobenz-2-oxa-1,3-diazol-4-yl)) was used as a model drug and encapsulated in the NPs. PLGA-PCB/NBD NPs were further conjugated with amine-modified galactose ligands and the resulting NPs with immobilized galactose ligands readily bound to the cells. Regrettably, no further work on this subject from their group has been reported as far as we know.

Du et al. do some work on PC based biodegradable drug carrier. They blended poly-ε-caprolactone (PCL) with PC-functionalized PCL (PC-PCL) to control the release of a model drug Ibuprofen (IB) and found that the release depends on degradation/erosion profiles and hydrophilicity of the blend matrices [71]. The IB release rate increased with the PC-PCL content because PC-PCL increased the hydrophilicity and biodegradability of the matrix.

2.3. Separation membrane

Membrane separation technology has been widely used for waste water treatment, municipal drinking water treatment, environmental engineering, and food-processing, etc., due to the high process efficiency, low energy consumption, easy scale-up and negligible environmental impact [72]. Generally, the separation membrane can be classified according to polymer substrate, like poly(vinylidene fluoride) (PVDF) [73], polyamide (PA) [74], poly(vinyl chloride) (PVC) [75], polypropylene (PP) [76], poly(ethersulfone) (PES) [77], polysulfone (Psf) [78], and polyacrylonitrile (PAN) [79]. However, the intrinsic

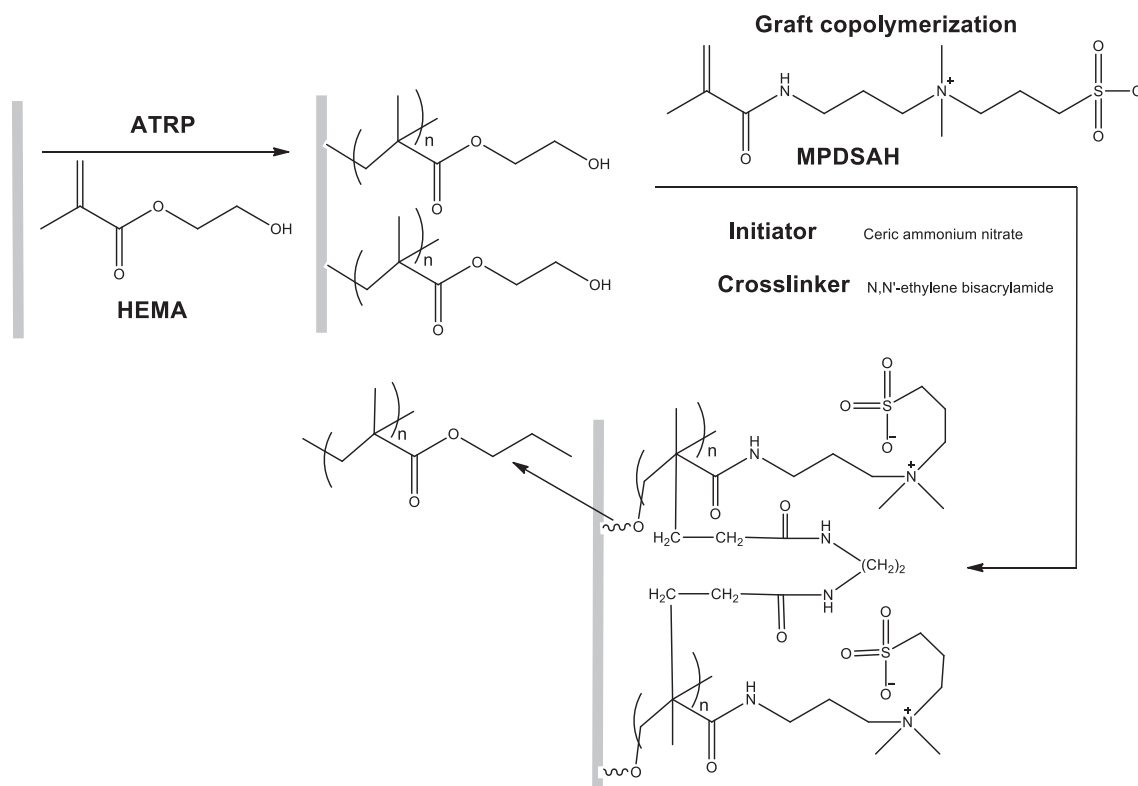


Fig. 6. Schematic diagram for the modification of PVDF membrane with sulfobetaine MPDSA via a two-step grafting polymerization [92].

hydrophobicity of these polymers results in biofouling in the filtration process [80]. The surfaces of membranes are easily polluted by the colloidal, organic, and other contaminant fouling [81], which results in evident decline in flux and increase in operational costs [82]. Since membrane performances, such as membrane fouling and water flux, are closely related with the chemical and physical properties of the outer layer of membranes [83], surface modification of membrane is an effective method to improve membrane performance. Zwitterionization is an effective method to modify the surface of separation membrane. It has been reported that as compared with a series of charged and neutral membranes with nearly identical pore size, the zwitterionic membranes showed best resistance property of protein adsorption [84].

Both zwitterionic monomers and polymers can be employed to modify the surface of separation membrane. Among zwitterions, SB and PSB are most commonly used because of their low cost and commercial availability. The surface of separation membrane should be pretreated before grafting zwitterionic polymers to address the inertness of polymer substrate and incompatibility between hydrophobic polymer and zwitterionic polymers, and different pretreatment and grafting methods should be employed for different polymer substrates due to the difference in surface character of polymer substrates. For example, UV-induced graft polymerization is employed to graft PSBMA and POEGMA [76] to the surface of PP. Chloromethylation is usually adopted as the first step to introduce $-\text{CH}_2\text{Cl}$ groups on PSf membrane surface, and the reactive groups are acting as substitution groups or surface-initiated ATRP (SI-ATRP) initiators for grafting PSBMA [78]. As far as PVC ultrafiltration membrane is regarded, a non-solvent-induced phase inversion process with in situ amination with triethylenetetramine and subsequent surface zwitterionization by sodium chloroacetate is employed [75]. While for PES, the surface should be pretreated before grafting zwitterionic polymers, or copolymers should be synthesized before blending, to address the inertness of PES and incompatibility between PES and zwitterionic polymers. For example, the surface of PES was pretreated with polydopamine (PDA). Then the zwitterionic copolymers were bound to the PES hollow fiber membranes via Michael

addition between $-\text{NH}_2$ on PDA and $-\text{SH}$ on zwitterionic random copolymers [77]. However, this ‘graft to’ method to incorporate zwitterionic polymer still encounters with potential problems such as less controllable coating density and reproducibility. In another case, ‘graft from’ was employed by Zhao et al. [85]. They synthesized polyethersulfone-*block*-poly(2-hydroxyethyl methacrylate) as the ‘glue’ to endow PES surface with reactive $-\text{OH}$ groups. $-\text{OH}$ was then converted into ATRP initiator to conduct the zwitterionic monomer polymerization. This ‘graft from’ method is easily to control zwitterionic polymer density by adjusting degree of polymerization. Because the zwitterionic polymer is chemically immobilized onto the PES-PHEMA layer, the coating stability mainly depends on the adhesion strength between amphiphilic PES-PHEMA copolymer and PES substrate. After 28 days of shaking in 40°C water, a slightly zwitterionic coating loss and water contact angle increased can be found. It shall be noted that the condition for stability test is not harsh and without any challenge of proteins or blood.

PVDF is a widely used membrane material due to its excellent mechanical properties, and thermal and chemical stability. But the intrinsic hydrophobicity of PVDF results in biofouling in the filtration process [80]. Therefore, PVDF is chosen as a representative to systematically discuss the modification of separation membrane with zwitterions.

PSB can be grafted onto the surface of PVDF membrane by physisorbed free radical grafting technique [86]. But PVDF membrane surface should be pretreated, for example, by atom ozone or alkaline activation, to provide active sites or groups for the following ATRP due to the inherent inertness of PVDF and incompatibility between highly polar PSBMA and hydrophobic PVDF membranes [87]. Regrettably, the ozone surface activation and subsequent graft polymerization of PVDF-g-PSB usually lead to low grafting yield and long modification time [88,89]. As shown in Fig. 6, alkaline activation of PVDF backbone can produce active $\text{C}=\text{C}$ functional group [90]. Then poly(*N,N*-dimethylamino-2-ethylmethacrylate) (MPDSA) was grafted onto PVDF backbones via ATRP, which further reacted with 1,3-propane sulfone to give

zwitterions. However, the alkaline treatment deteriorated mechanical strength of PVDF membrane seriously [91]. In order to address this problem, poly(2-hydroxyethyl methacrylate) (PHEMA) was grafted from the outside surface of PVDF hollow fiber membrane through ATRP before grafting 3-(methacryloylamino) propyl-dimethyl-(3-sulfopropyl) ammonium hydroxide (MPDSA) to maintain the mechanical performance of PVDF membrane and improve the anti-fouling property [92,93]. The modified PVDF membrane shows good anti-fouling capacity with relative flux recovery achieving 98.2% and a low irreversible fouling extent which is lower 10 times than that of the unmodified PVDF hollow fiber membrane, and oil-fouling-resistance property with a relative flux recovery of about twice that of the native PVDF membrane. It should be mentioned that amide group of sulfobetaine played an important role in the long-term stability of the modified PVDF membrane [94]. The poly(2-(methacryloyloxyethyl) ethyldimethyl-(3-sulfopropyl) ammonium)-g-PVDF membrane lost the grafted polymers after cleaned in membrane cleaning solution for 12 days, while the poly(methacryloylamino) propyl-dimethyl-(3-sulfopropyl) ammonium hydroxide-g-PVDF membrane could maintain without variation due to the presence of the amide group in the grafted layer on PVDF membrane. Therefore, PSB with amide group has great application value for the modification of separation membranes.

Besides PSB, PCB has also been used to modify PVDF, though less commonly. A PCBMA layer was created on PVDF membrane surface via physisorbed free radical polymerization grafting technique [95]. Grafted density of PCBMA could be easily controlled by reaction time and monomer concentration. The water flux recovery ratio and protein antifouling ability of modified PVDF membrane improved significantly.

Recently, separation membrane integrating antifouling property and self-cleaning performance attracts great interest of researchers. The self-cleaning surfaces often have nonpolar hydrophobic polymers such as silicone-based [97] and fluorinated [98] polymers featured with low surface energy. Unlike hydrophilic surfaces that can resist the adsorption of foulants, the hydrophobic self-cleaning surfaces minimize the interaction between foulants and surface, so that foulants can be easily cleaned and removed by low hydrodynamic shear forces or by simple mechanical forces [99,100]. Jiang et al. synthesized amphiphilic copolymer composing of zwitterionic poly[3-(methacryloyl-amino) propyl-dimethyl(3-sulfopropyl) ammonium hydroxide] and low surface energy poly(hexafluorobutylmethacrylate) via potassium persulfate (KPS) initiated aqueous phase emulsion polymerization based on zwitterionic monomer and side-chain fluorinated methacrylate monomer with sodium dodecylsulfate (SDS) as emulsifier (Fig. 7) [66]. The resulting copolymer was used as an additive to blend with PVDF membranes by the phase-inversion process. Since the membrane surfaces were covered with both zwitterionic and low surface energy polymer brushes, the membrane possessed superior antifouling and self-cleaning properties. The flux-decline was reduced to an ultralow level

and the flux recovery was close to 100% for both oil/water emulsion separation as well as bioseparation. Considering their superior combined properties, separation membrane integrating antifouling property and self-cleaning performance may be the main direction of separation membrane.

2.4. Marine coating

Marine biofouling is a global problem for maritime industries with severe economic and environmental penalties which is the undesired growth of microorganisms, barnacles and seaweeds on any submerged surfaces [101]. One study show that marine biofouling comprises 30% of the ships in the US Navy fleet, hull fouling costs about \$56 million per annum [102]. There are three well-established strategies to combat marine biofouling: toxic coatings, fouling release (FR) coatings and nonfouling/antifouling coatings. Traditional method is to use toxic substances such as tin based paints which deter the living creatures from adhering to a surface. After tin paints are banned, copper based paints is being used now. Copper based paints are getting banned world over. For example, San Diego Bay had dissolved copper concentrations almost 4 times the limit permissible [103]. Next type of coating is, fouling release (FR) coating, which is made of elastic coatings such as poly(dimethylsiloxane) (PDMSe). It releases foulants settling on the surface by the force generated by water flow on the surface. The PDMSe FR coatings have several drawbacks: a) hydrophobic, so prone to more fouling; b) difficult to apply on surface due to their low surface energy; c) mechanically less durable. Furthermore, PDMSe frequently 'fails' to diatoms [104,105].

Recent research mainly focuses on the development of biocide-free "Green" alternatives which approach the problem of fouling through the physio-chemical and bulk materials properties of surfaces. Employment of amphiphilic polymers either by crosslinked themselves or coated on top of a soft elastomeric base can both lower the fouling and enable fouling release [106,107]. Most of the amphiphilic or hydrophilic coatings to combat marine biofouling are based on PEG. As mentioned before, PEG is unstable and prone to undergo degradation due to the presence of ether bonds, while zwitterionic polymers are more chemically stable and most effective for nonfouling applications as they bind to water through ionic interactions and thus show electrostatically induced hydration in comparison to hydrogen bonding based interactions of PEG [108].

PSBMA brush was grafted on glass surface and tested against marine fouling of marine green seaweed *Ulva* and the diatom *Navicula* [109]. It was found that the zwitterionic polymers completely inhibited the settlement of spores of the green *Ulva* and the attachment of diatom cells. This is the first paper to demonstrate the biofouling inhibiting effect of tethered zwitterionic coatings for marine species. To explore the effectiveness of resistance of zwitterionic polymers towards other

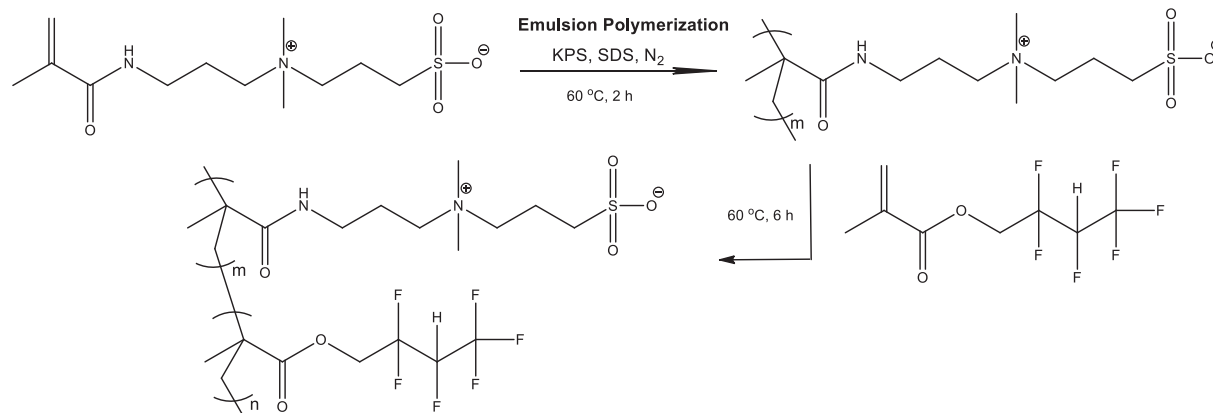


Fig. 7. Synthesis of amphiphilic copolymers [96].

marine organisms, a study was conducted using cypris larvae barnacles on zwitterionic polymer surfaces. The movement of the cypris larvae was tracked in the study using Noldus Ethovision 3.1 [110]. Two types of zwitterionic polymers, PSBMA (a sulfobetaine) and PCBMA (a carboxybetaine) were tested in this study. Both the zwitterionic polymer surfaces showed excellent resistance as the larvae was unwilling to settle on the surface. On PSBMA, cyprids explored the surface and unwilling or unable to settle, whereas on PCBMA cyprids did not attempt exploration and left the surface quickly. In another study, anti-fouling character of zwitterionic polymer brushes against marine organisms was investigated by performing settlement tests with barnacle cypris larvae and mussel larvae in their adhesion period [111]. Barnacle attachment/detachment studies were performed on sulfobetaine zwitterionic polymer brushes. In this study the PSBMA brushes were thin (1–2 μm) and thus antifouling purely relies on the surface chemistry. These surfaces were able to resist the attachment of the barnacles after allowing them to stick on the surface for 14 days [112]. Marine bacteria colonization test was also performed. These studies highlight the potential of the zwitterionic polymers as fouling-resistant surface for marine applications.

Besides the main applications discussed above, zwitterionic polymeric materials can be used in various other fields, such as template and catalyst, and dye absorption. For instance, zwitterionic microgel prepared from 2-(methacryloyloxy) ethyl dimethyl (3-sulfoethyl) ammonium hydroxide is used as a template for the synthesis of Ni nanoparticles. The prepared Ni composites were shown to have very high catalytic performance in the reduction of nitro aromatic compounds [113]. Sahiner et al. prepared zwitterionic microgels via simple ring opening alkylation reaction of PEI with 1,3-propane sultone. It was demonstrated that the resulting microgels can be used as versatile adsorbent for the removal of toxic anionic or cationic dyes due the zwitterionic character and the modified PEI shows good biocompatibility [114].

3. Conclusions

Biofouling is one of the serious problems in biomedicine, chemical separation membrane to marine coating. Therefore, the development of anti-fouling materials is extremely significant. Zwitterionic polymers are highly hydrophilic and no doubt crucial for a variety of antifouling applications. By far, much progress has been made from their application in biomedicine, chemical separation membrane to marine coating. The design flexibility, and ability of functionalization and hydrolysis make PCB and its derivatives useful for next-generation NPs and blood contacted sensors. The specific recognition of C-reactive protein, cell-membrane-penetration abilities, and lubrication properties extend the application of PMPC for C-reactive protein inhibitors, prodrug carriers, and joint replacement. The low cost and commercial availability make SB and PSB most promising to be industrialized in separation membrane and marine coating.

However, to date, the applications of zwitterionic materials are still at a laboratory scale, and many difficulties encountered in the industrial application. There is still much work to do and a long way to go to realize the industrial application of these zwitterionic materials. Biodegradability is required for certain biomedical applications, while current zwitterionic polymers mostly have nondegradable carbon-carbon backbones. Future work should make more effort on biodegradable zwitterionic materials/devices with superior antifouling properties. The cost of zwitterionic monomer and polymer synthesis and modification for the surface is high. Researchers and engineers will need to reduce the synthesis and modification cost, and improve the yield and thermal stability by designing the optimal means. For drug deliveries, future work should aim at developing biocompatible, low-toxic, nonimmunogenic, long-circulating, structurally regulable, multifunctional and targeting drug delivery system with slow-releasing rate, and all obstacles encountered in vitro clinical applications need to be

considered. For the biosensor and separation membrane, it should aim at detecting and separating several substances simultaneously. More effort is needed to improve the endurance of separation membrane and marine coating, cut down the cost and simplify the preparation process. More work should be focused to broaden applications of zwitterionic materials.

Zwitterionic materials will make great contribution in biomedical devices, implants and drug deliveries, separation membranes and marine coating.

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